

TE Curve Verification Pipeline Selected Active Model Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates only the currently selected active model families against the repository held-out TE-curve test split. The `global` surface is intentionally excluded from this reduced decision report.

Dataset And Split

- dataset config: `config/datasets/transmission_error_dataset.yaml` ;
- dataset root: `data\polished_dataset` ;
- comparison mode: `selected_active_model_matrix` ;
- candidate count: `7` ;
- held-out curve count before candidate filtering: `94` ;
- percentage-error denominator: `peak_to_peak_truth` ;
- evaluated surface: `Bw` ;
- evaluated direction: `backward` ;

Candidate Inventory

Candidate	Source	Family	Surface	Valid Directions
<code>polished_actual_values_feedforward_Bw</code>	<code>polished_actual_values_model_archive</code>	<code>feedforward</code>	<code>Bw</code>	<code>backward</code>
<code>polished_actual_values_tree_Bw</code>	<code>polished_actual_values_model_archive</code>	<code>tree</code>	<code>Bw</code>	<code>backward</code>
<code>polished_actual_values_harmonic_regression_Bw</code>	<code>polished_actual_values_model_archive</code>	<code>harmonic_regression</code>	<code>Bw</code>	<code>backward</code>
<code>polished_actual_values_periodic_mlp_harmonic_Bw</code>	<code>polished_actual_values_model_archive</code>	<code>periodic_mlp_harmonic</code>	<code>Bw</code>	<code>backward</code>
<code>polished_actual_values_periodic_gru_sequence_Bw</code>	<code>polished_actual_values_model_archive</code>	<code>periodic_gru_sequence</code>	<code>Bw</code>	<code>backward</code>
<code>polished_actual_values_wave4_1_mae_robust_loss_Bw</code>	<code>polished_actual_values_model_archive</code>	<code>wave4_1_mae_robust_loss</code>	<code>Bw</code>	<code>backward</code>
<code>polished_actual_values_wave4_2_quantile_p10_p50_p90_Bw</code>	<code>polished_actual_values_model_archive</code>	<code>wave4_2_quantile_p10_p50_p90</code>	<code>Bw</code>	<code>backward</code>

Exact Model Paths

Candidate	Family	Surface	ONNX Model Path	Python Model Path
polished_actual_values_feedforward_Bw	feedforward	Bw	models/polished_dataset/actual_values/exported/feedforward/backward/2026-07-07-19-17-09_te_feedforward_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/feedforward/backward/2026-07-07-19-17-09_te_feedforward_bw_polished_actual_values/python/feedforward-epoch=074-val_mae=0.00164741.ckpt
polished_actual_values_tree_Bw	tree	Bw	models/polished_dataset/actual_values/exported/tree/backward/2026-07-07-09-58-04_te_tree_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/tree/backward/2026-07-07-09-58-04_te_tree_bw_polished_actual_values/python/tree_model.pkl
polished_actual_values_harmonic_regression_Bw	harmonic_regression	Bw	models/polished_dataset/actual_values/exported/harmonic_regression/backward/2026-07-08-00-32-21_te_harmonic_regression_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/harmonic_regression/backward/2026-07-08-00-32-21_te_harmonic_regression_bw_polished_actual_values/python/harmonic_regression-epoch=054-val_mae=0.00182643.ckpt
polished_actual_values_periodic_mlp_harmonic_Bw	periodic_mlp_harmonic	Bw	models/polished_dataset/actual_values/exported/periodic_mlp_harmonic/backward/2026-07-08-03-31-01_te_periodic_mlp_harmonic_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/periodic_mlp_harmonic/backward/2026-07-08-03-31-01_te_periodic_mlp_harmonic_bw_polished_actual_values/python/periodic_mlp-epoch=128-val_mae=0.00117146.ckpt
polished_actual_values_periodic_gru_sequence_Bw	periodic_gru_sequence	Bw	models/polished_dataset/actual_values/exported/periodic_gru_sequence/backward/2026-07-09-01-09-21_te_periodic_gru_sequence_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/periodic_gru_sequence/backward/2026-07-09-01-09-21_te_periodic_gru_sequence_bw_polished_actual_values/python/periodic_gru_sequence-epoch=257-val_mae=0.00127934.ckpt
polished_actual_values_wave4_1_mae_robust_loss_Bw	wave4_1_mae_robust_loss	Bw	models/polished_dataset/actual_values/exported/wave4_1_mae_robust_loss/backward/2026-07-12-19-08-24_te_wave4_1_mae_robust_loss_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_1_mae_robust_loss/backward/2026-07-12-19-08-24_te_wave4_1_mae_robust_loss_bw_polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=173-val_mae=0.00178689.ckpt
polished_actual_values_wave4_2_quantile_p10_p50_p90_Bw	wave4_2_quantile_p10_p50_p90	Bw	models/polished_dataset/actual_values/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-17-11-13_te_wave4_2_quantile_p10_p50_p90_bw_polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-17-11-13_te_wave4_2_quantile_p10_p50_p90_bw_polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=123-val_mae=0.00178818.ckpt

Metric Ranking

Rank	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
1	polished_actual_values_periodic_gru_sequence_Bw	0.001333	0.001672	2.625	5.438
2	polished_actual_values_wave4_2_quantile_p10_p50_p90_Bw	0.002188	0.002651	3.867	10.067
3	polished_actual_values_wave4_1_mae_robust_loss_Bw	0.002315	0.002784	3.989	10.342
4	polished_actual_values_periodic_mlp_harmonic_Bw	0.002483	0.002914	4.321	10.519
5	polished_actual_values_feedforward_Bw	0.002769	0.003293	4.919	10.603
6	polished_actual_values_tree_Bw	0.002759	0.003293	4.935	10.731

Rank	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
7	polished_actual_values_harmonic_regression_Bw	0.002966	0.003511	5.369	10.974

Direction Breakdown

Direction	Candidate	Curve MAE [deg]	Curve RMSE [deg]	Mean Percentage Error [%]	P95 Mean Percentage Error [%]
backward	polished_actual_values_periodic_gru_sequence_Bw	0.001333	0.001672	2.625	5.438
backward	polished_actual_values_wave4_2_quantile_p10_p50_p90_Bw	0.002188	0.002651	3.867	10.067
backward	polished_actual_values_wave4_1_mae_robust_loss_Bw	0.002315	0.002784	3.989	10.342
backward	polished_actual_values_periodic_mlp_harmonic_Bw	0.002483	0.002914	4.321	10.519
backward	polished_actual_values_feedforward_Bw	0.002769	0.003293	4.919	10.603
backward	polished_actual_values_tree_Bw	0.002759	0.003293	4.935	10.731
backward	polished_actual_values_harmonic_regression_Bw	0.002966	0.003511	5.369	10.974

Curve Evidence

Each selected candidate is shown with the same four deterministic held-out operating conditions for this direction. The dark line is the measured TE curve and the blue line is the model prediction for the same operating condition.

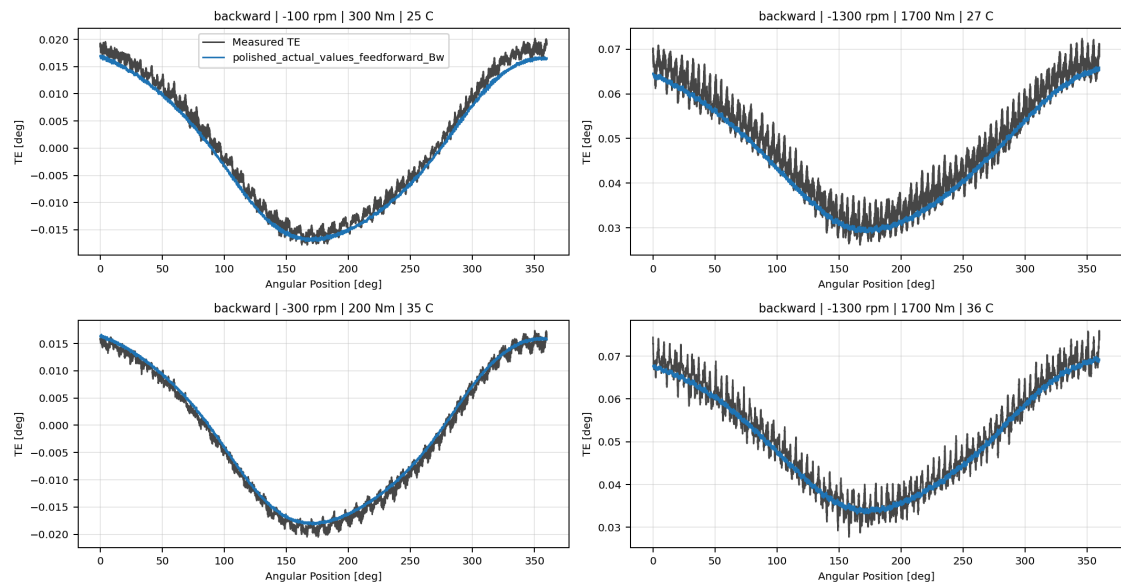
Shared operating conditions

- 100 rpm, 300 Nm, 25 C
- 1300 rpm, 1700 Nm, 25 C
- 300 rpm, 200 Nm, 35 C
- 1300 rpm, 1700 Nm, 35 C

Candidate	Role	Curve MAE [deg]	Curve RMSE [deg]	Mean MPE [%]	P95 MPE [%]
polished_actual_values_feedforward_Bw	Selected candidate	0.002769	0.003293	4.919	10.603
polished_actual_values_tree_Bw	Selected candidate	0.002759	0.003293	4.935	10.731
polished_actual_values_harmonic_regression_Bw	Selected candidate	0.002966	0.003511	5.369	10.974
polished_actual_values_periodic_mlp_harmonic_Bw	Selected candidate	0.002483	0.002914	4.321	10.519
polished_actual_values_periodic_gru_sequence_Bw	Surface leader	0.001333	0.001672	2.625	5.438
polished_actual_values_wave4_1_mae_robust_loss_Bw	Selected candidate	0.002315	0.002784	3.989	10.342
polished_actual_values_wave4_2_quantile_p10_p50_p90_Bw	Selected candidate	0.002188	0.002651	3.867	10.067

polished_actual_values_feedforward_Bw

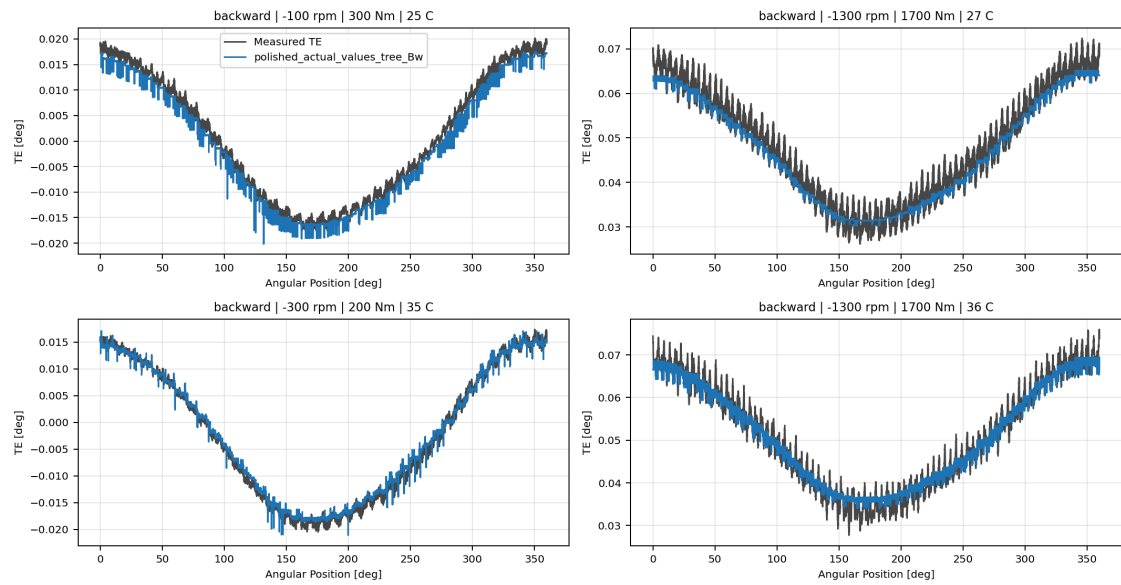
polished_actual_values_feedforward_Bw



polished_actual_values_feedforward_Bw measured-versus-predicted TE collage

polished_actual_values_tree_Bw

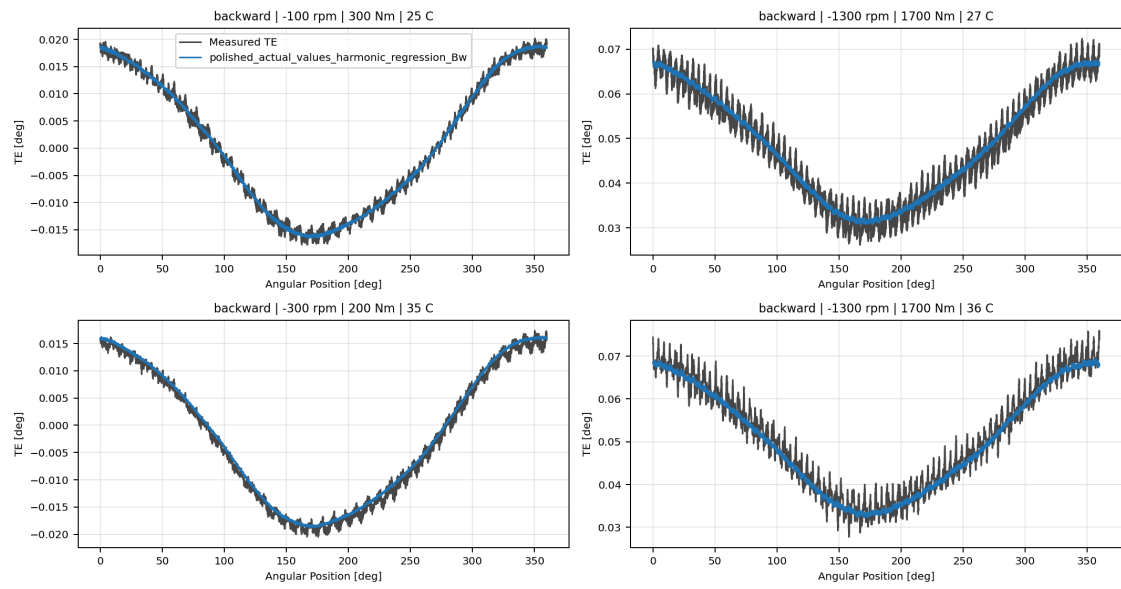
polished_actual_values_tree_Bw



polished_actual_values_tree_Bw measured-versus-predicted TE collage

polished_actual_values_harmonic_regression_Bw

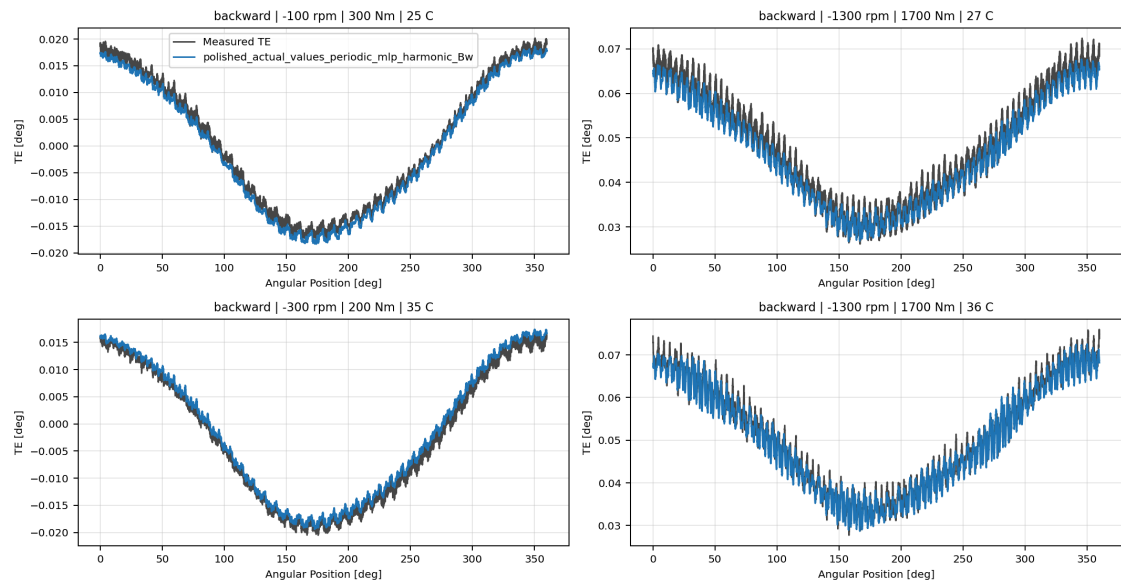
polished_actual_values_harmonic_regression_Bw



polished_actual_values_harmonic_regression_Bw measured-versus-predicted TE collage

polished_actual_values_periodic_mlp_harmonic_Bw

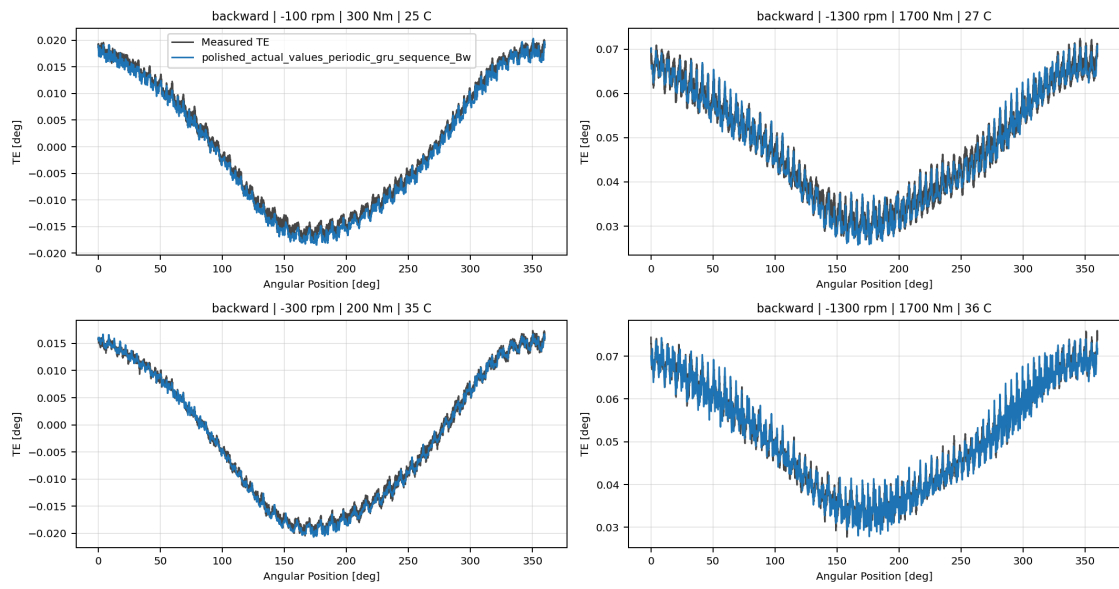
polished_actual_values_periodic_mlp_harmonic_Bw



polished_actual_values_periodic_mlp_harmonic_Bw measured-versus-predicted TE collage

polished_actual_values_periodic_gru_sequence_Bw

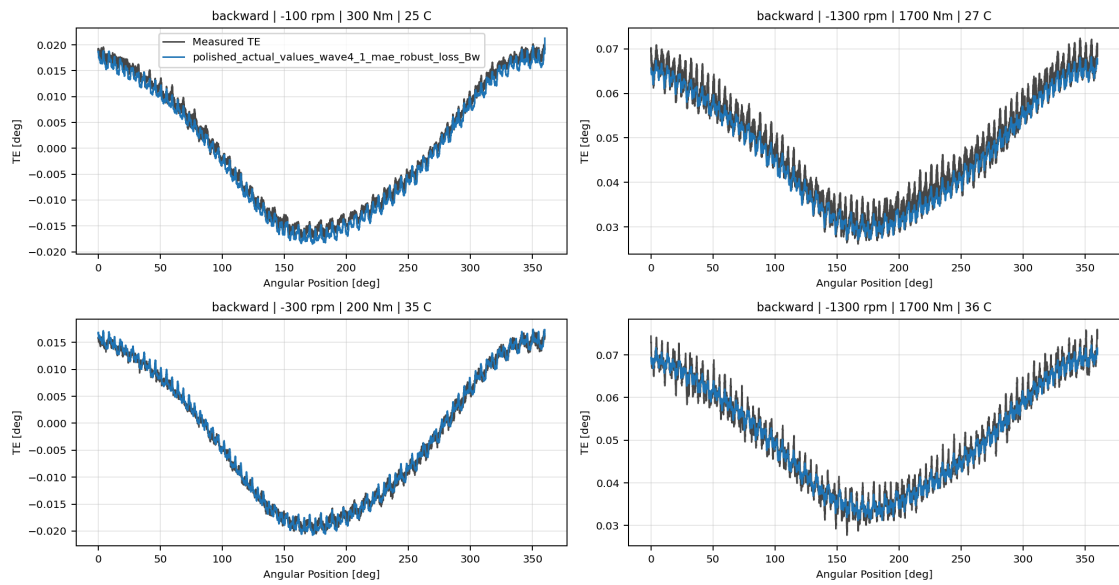
polished_actual_values_periodic_gru_sequence_Bw



polished_actual_values_periodic_gru_sequence_Bw measured-versus-predicted TE collage

polished_actual_values_wave4_1_mae_robust_loss_Bw

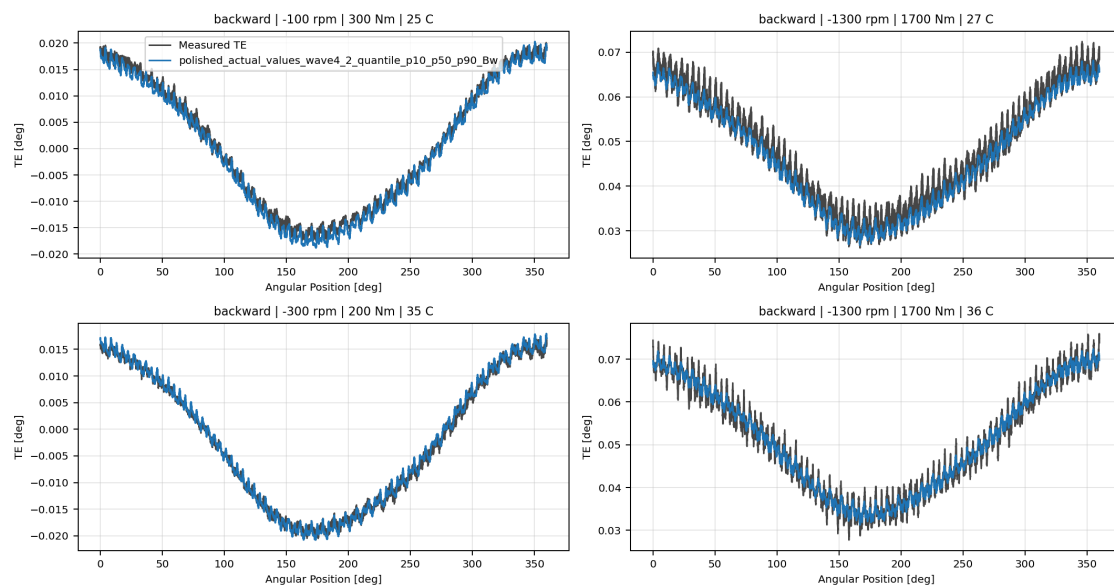
polished_actual_values_wave4_1_mae_robust_loss_Bw



polished_actual_values_wave4_1_mae_robust_loss_Bw measured-versus-predicted TE collage

polished_actual_values_wave4_2_quantile_p10_p50_p90_Bw

polished_actual_values_wave4_2_quantile_p10_p50_p90_Bw



polished_actual_values_wave4_2_quantile_p10_p50_p90_Bw measured-versus-predicted TE collage

Artifacts

- summary YAML: `output\validation_checks\track2_reference_comparison\2026-07-19-17-49-24_track2_selected_active_polished_actual_values_matrix_track2_selected_active_polished_actual_values_backward_2026_07_19\validation_summary.yaml` ;
- per-condition CSV: `output\validation_checks\track2_reference_comparison\2026-07-19-17-49-24_track2_selected_active_polished_actual_values_matrix_track2_selected_active_polished_actual_values_backward_2026_07_19\per_condition_metrics.csv` ;
- grouped report plot root: `doc\reports\campaign_results\track_2\verification_plots\selected_active_model_matrix` ;
- grouped report plot count: `0` ;

Interpretation

Rows are ranked by mean percentage error within this selected-active surface report. The curve-evidence section uses shared deterministic held-out operating conditions so visual shape fidelity can be compared across the selected families.

Open Gaps

- This remains an offline TE-curve comparison and does not replace the future online `Table 9` compensation benchmark.
- RCIM Model-Bank Reproduction remains a separate paper-reference benchmark path and is not part of this selected-active model-only report.